

Interpretable power load forecasting model based on CNN GRU and SHAP

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Abstract—Accurate power load forecasting is very important for the rational planning and operation scheduling of power grid. While AI technology has excelled at load forecasting, it remains a challenging problem to explain why these inaccurate results occur when the forecasts are inaccurate. Interpretable artificial intelligence(XAI) has recently received quite a bit of attention in solving this problem. In order to mine the temporal characteristics in the data, improve the accuracy of load prediction, and better explain the black box model, a power load prediction model based on convolutional neural network(CNN) and gated cyclic unit(g) hybrid neural network was established, and Shapley addition interpretation was used to analyze and explain the model. In this paper, we use the data of an electric power competition in 2016 to conduct experiments, input the cleaned data into the proposed model for training, and compare it with the three baseline models of LSTM, GRU and CNN LSTM to verify the practicability and effectiveness of the proposed method. Finally, SHAP is applied to the CNN GRU model to improve the interpretability of the model.

Keywords—power load forecasting; deep learning; convolutional neural network; gated recurrent unit; explainable artificial intelligence

I. INTRODUCTION

In recent years, with the development of smart grid, power load forecasting is not only important for the stability of the grid system, maximizing economic benefits to provide protection, but also important for the rational planning and arrangement of the power grid^[1]. In order to meet the power demand^[2], ensure power security and improve the efficiency of power consumption to reduce energy consumption, so it is imperative to improve the accuracy of power load forecasting.

At present, the methods of power load forecasting are mainly categorized into two types which are traditional mathematical and statistical methods and machine learning methods. The traditional statistical methods are efficient but require high data smoothness for time series. In machine learning methods, although they have the ability to deal with nonlinear and complex problems, they cannot adequately capture the time series characteristics.

In recent years, in the rapid development and wide application of deep learning, convolutional neural networks (CNN), long and short-term memory networks (LSTM), and gated recurrent units (GRU) have attracted great attention in the field of load forecasting. CNN can capture seasonal cycles and accurately predict nonlinear sequences, but the accuracy decreases in the case of large fluctuations^[3]. LSTM have a better capability of processing time series data, but the model

convergence is slow^[4]. GRU simplifies the internal structure of LSTM and shortens the model training time while maintaining high prediction accuracy, but it cannot fully exploit the relationship of discontinuous features^[5].

Despite the rapid development of AI technology, it is not easy to apply it in the industrial field^[6]. In addition, AI techniques use a longitudinal learning structure through which results are obtained from the provided data, but this method seems to present a black-box state that makes it difficult to understand the internal decision-making process^[7]. Recently, explainable AI (XAI) techniques have attracted considerable attention in addressing this problem. The research direction of this technique is divided into two main categories^[8], the first category is to propose a new AI model which are able to provide an explainable decision process when making a decision, i.e., the model explains why a particular decision was made, making the decision more understandable. The second category is to add descriptive labels or explanations based on the outputs of existing AI models and transform these labels so that the user can understand the model's decision-making process. Both approaches aim to improve the explainability of AI systems and thus increase trust in the model's decisions.

This paper proposes an interpretable CNN-GRU load forecasting model based on the above research. The method takes the date factor, meteorological factor, and historical load factor as the inputs to the model, first extracts the features of the normalized data efficiently using the CNN network and retains the features of the original dataset, and then inputs the extracted features into the GRU network for training to get the results of the load prediction, and finally explains the prediction model using the using the Shapley additive interpretation (SHAP). The experimental results show that the method has higher prediction accuracy and the model is well interpreted.

II. LOAD INFLUENCING FACTORS

When constructing a forecasting model, a reasonable choice of forecasting feature set is crucial to ensure the convergence of the model and improve the forecasting accuracy. In this paper, a priori knowledge of load forecasting is fully considered and the load influencing factors are integrated into the input parameters. The influencing factors are categorized into external and internal factors. External factors include time stamps and weather information^[9]. The main variable

considered in internal factors is historical electricity consumption^[10].

A. Timestamp Information

Time-stamped information, such as month, hour, week, and holiday indicators, can present different patterns of electricity usage^[11]. Hourly electricity usage can vary depending on work hours, meal times, and other factors. Different days of the week and holidays can also have an impact on the electricity consumption pattern, presenting different electricity demand patterns. different weekdays and holidays. In addition, in order to reflect the impact of different weekdays and holidays on electricity consumption patterns, we include the holiday indicator as an input variable. At the same time, legal national holidays are added to the holiday indicator as well.

B. Weather Information

Weather information plays a very important role in load forecasting because the frequency of energy intensive appliances such as air conditioners and heaters is closely related to the weather. Weather information includes probability of precipitation, amount of precipitation, daily maximum and minimum temperatures, wind speed and direction, and humidity. This information is important to understand how weather conditions affect electricity loads because they can help forecasting models predict changes in electricity demand more accurately^[12].

In order to increase the applicability of the method in practical applications, we use average temperature, humidity and rainfall as input variables, as they are closely related to electricity consumption.

C. Historical Electricity Consumption

Historical electricity consumption data are considered as an internal factor because they reflect recent trends in electricity consumption^[13]. To take this factor into account, we will use electricity consumption data from the day before and the week before the forecast time point. The electricity consumption of the previous day reflects the most recent trend, while the electricity consumption of the previous week reflects the long-term trend of the day. Given the potential differences in electricity consumption patterns between weekdays and holidays, we will consider them separately to more accurately forecast future electricity demand. That is, holiday indicators are added to indicate different electricity consumption patterns.

In summary, this paper constructs an 11-dimensional feature vector containing timestamp information, weather information, and historical electricity consumption as inputs to the prediction model, as shown in Table 1.

III. MODEL FUNDAMENTALS

A. Convolutional Neural Network

Convolutional neural network CNN model, is a popular model in the field of deep learning and one of the classical algorithms that are widely used^[14]. It usually consists of three main components which are convolutional layer, pooling layer and fully connected layer. The convolutional layer utilizes

multiple convolutional kernels to perform convolutional operations on the input data in order to extract features, and then enhances the feature representation by nonlinear mapping through activation functions. The pooling layer implements the aggregation of features obtained from convolutional operations through operations such as maximum pooling and mean pooling to reduce the dimensionality of the data while retaining the most important feature information. This helps to reduce the computational burden and improve the robustness of the model. The fully connected layer sits at the top of the network and integrates the features extracted by the convolution and pooling layers, which are then passed to the output layer to compute the final classification or regression results. These three components work in tandem to process the input data and provide the final classification or regression results.

Table 1: Input variables and their information for modeling

Influencing Factors	Characterization Name	Characterization Description
Timestamp Information	dayofmonth	1~12 represents January to December
	dayofweek	1~7 represents Monday to Sunday
	hourofday	0~23 represents 1~24 o'clock
	holiday	0 for non-holiday, 1 for holiday
	work	0 for non-working hours, 1 for working hours
Weather Information	tem	Average temperature per day
	hun	Humidity per day
	mm	Rainfall per day
Historical Electricity Consumption	lag_1	Current load of the previous day
	Lag_7	Current load of the previous week

B. Gated Recurrent Unit

Recurrent Neural Networks (RNN) are prone to the problem of gradient vanishing and explosion. Long Short-Term Memory Networks (LSTM) and Gated Recurrent Units (GRU), as improved network types of RNNs, then solve this problem well.

Although LSTM is able to capture long-term dependencies when dealing with time series data, its complex internal structure leads to a long training time. GRU is an optimized improvement of LSTM, which simplifies its internal structure while guaranteeing the prediction accuracy.

The LSTM architecture consists of three gates, an input gate, a forgetting gate, and an output gate. On the other hand, GRU^[15] combines the input and forgetting gates in the LSTM into a single update gate and retains the reset gate, an improvement that makes the GRU structurally more simplified while maintaining its performance in sequential data tasks. The basic structure of the GRU network is shown in Figure 1.

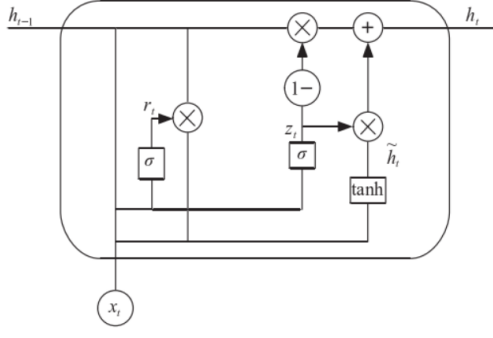


Fig. 1. Structure of gated recurrent unit

Its calculation formula is as follows:

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \quad (1)$$

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \quad (2)$$

$$\tilde{h}_t = \phi(W_{\tilde{h}} \cdot [r_t \times h_{t-1}, x_t]) \quad (3)$$

$$h_t = (I - z_t) \times h_{t-1} + z_t \times \tilde{h}_t \quad (4)$$

$$y_t = \sigma(W_o \cdot h_t) \quad (5)$$

the input is denoted by x_t , the state memory variable of the previous moment is h_{t-1} , and the state memory variable of the current moment is h_t . r_t is the update state of the gate, z_t is the reset state of the gate, and $\tilde{h}_t$ indicates the real-time candidate state, y_t is the current moment output. W_r , W_z , $W_{\tilde{h}}$ and W_o are the weight parameter matrices of update gate, reset gate, candidate set, and output vector, respectively. σ represents the sigmoid activation function and ϕ represents the tanh activation function.

C. CNN-GRU Network Hybrid Model

The hybrid CNN-GRU network model proposed in this article takes time series feature maps as input. These features include timestamp information, weather data, and historical load information. The historical load value at each time point is collectively represented by various characteristics associated with that moment. The input time series data is then gradually transformed into feature maps using a sliding window approach, which are subsequently fed into the CNN-GRU hybrid model. The model is depicted in Figure 2. In this study, we first employ the convolutional layer and the pooling layer to extract features from the time series data. Next, we input these extracted features into the GRU layer for training, and ultimately, we generate prediction results through a fully connected layer.

D. Model Interpretability

In complex models of artificial intelligence, it is difficult to accurately quantify the extent to which each input variable affects the final outcome^[16]. Therefore, it is necessary to adopt a strategy to better understand the behavior of the model by approximating the output values of the original model to those

of the interpretable model. Furthermore, SHAP is a tool that uses Shapley values to assess the correlation between independent and dependent variables in model learning^[17].

In addition, SHAP uses conditional expected value functions to learn the Shapley values of the model, and includes kernel SHAP for all machine learning and deep SHAP for neural networks. In this paper, deep SHAP is used to calculate the SHAP values of all independent variables in the data set, and the importance of variables and partial dependency graph are visualized.

At present, substitution method is used to measure the importance of a feature in the model^[18]. Although this method is fast to calculate, the results can be misinterpreted when the independent variables are interdependent, and the negative effects of the independent variables cannot be calculated. The SHAP method, on the other hand, takes into account the possibility of interaction between variables and quantifies their negative effects. Therefore, although it has the disadvantage of being slow, it can assess the impact more accurately than the replacement method.

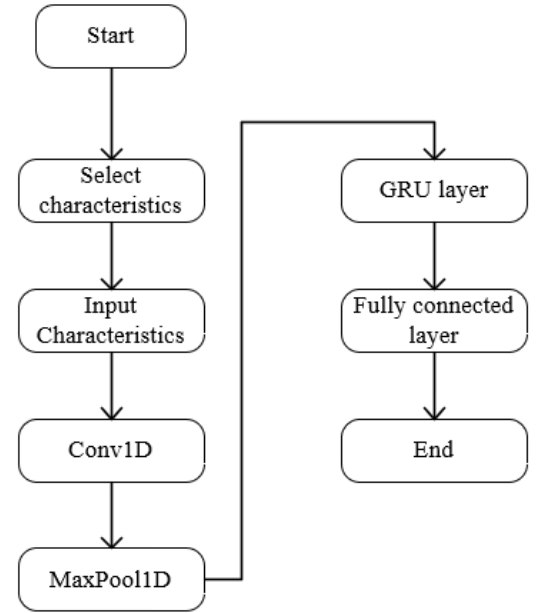


Fig.2Structure of CNN-GRU

IV. EXPERIMENTAL RESULT

A. Experimental Data And Environment

The experimental data is sourced from the 2016 electric power competition dataset, which is divided into two regions. For this experiment, the data from the second region is selected. The dataset is further split into training, validation, and test sets in an 8:1:1 ratio.

The operating system is Windows 10, the CPU is Intel(R) Core(TM) i5-8300H, the graphics card is NVIDIA GeForce GTX 1060, the development language is Python 3.7, and the framework is PyTorch.

B. Experimental evaluation criteria

To compare model performance, we use the mean absolute percentage error (MAPE), mean absolute error (MAE), and root mean square error (RMSE) as evaluation metrics. These metrics are calculated using the following formulas:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|\tilde{y}_i - y_i|}{y_i} \times 100 \quad (6)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\tilde{y}_i - y_i| \quad (7)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - y_i)^2} \quad (8)$$

N is the number of predicted time points, i is the predicted time point, $\tilde{y}_i$ is the predicted time value, and y_i is the actual time value.

C. Analysis of Experimental Results

This experiment started with a stepwise model tuning using the control variable method. The prediction effect played by increasing the depth of the model is tested by continuously increasing the number of layers of the GRU network. The effect of different number of GRU network layers on the prediction results is tested under the condition that the extracted features of fixed CNN layers remain unchanged. The test results are shown in Table 2, and it can be found that the training effect is best when the GRU network layer is 1.

Table 2: Experimental results of model structure combination

GRU layers	Epoch	Feature	MAPE
1	200	10	2.23
2	200	10	2.35
3	200	10	2.56
4	200	10	2.97

After the above experiments, the parameters of the model and the model structure are determined, the CNN-GRU model is trained and tested. And it is compared with the baseline model on the test set. The comparison of MAPE value, MAE value, and RMSE value of the prediction results are shown in Table III. And the predicted and actual values of each model are taken at the last 24 moment points on the test set and plotted as shown in Figure 3.

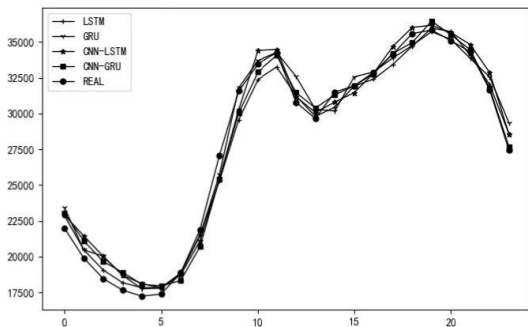


Fig.3 Comparison graph of prediction results of different models

As can be seen from Fig. 3, the CNN-GRU hybrid network model proposed in this paper predicts the load curve with the highest accuracy, which is basically consistent with the actual load change trend, and has a good prediction accuracy.

From Table 3, it can be seen that the CNN-GRU model proposed in this paper has the smallest mean MAE and mean RMSE and the highest prediction accuracy MAPE. Compared with the traditional LSTM and GRU networks, the prediction accuracy is improved by 0.43 and 0.53, the MAE is decreased by 131.56 and 151.57, and the RMSE is decreased by 129.82 and 229.59, respectively. The prediction accuracy of the CNN-LSTM model is slightly improved by a small amount, but the training time of the model can be shortened by 18%, which improves the prediction efficiency.

Table 3: Comparison of model prediction accuracy

Model	MAPE	MAE	RMSE
LSTM	2.35	659.31	796.39
GRU	2.45	697.32	896.16
CNN-LSTM	2.05	559.07	729.49
CNN-GRU	1.92	527.75	666.57

Table 4 Comparison of model prediction accuracy

Model	Train time/s	Test time/s
CNN-LSTM	241.49	2.80
CNN-GRU	203.63	3.34

D. Model Interpretation

Explanatory methods are a crucial strategy to ensure a model's reliability and are currently receiving significant attention. SHAP is a method used to explain predictions based not only on the importance of features in a complex training model but also on the predictions for the current test samples. In this paper, the DEEP SHAP method is employed to enhance the interpretability of CNN-GRU-based models by quantifying the contribution of each input variable to the prediction results. Additionally, this paper utilizes summary plots and correlation plots to help elucidate the decision-making process and prediction results of the model in a more intuitive and understandable manner.

The overall feature summary analysis was performed in the SHAP framework, and the CNN-GRU model variable importance rankings as well as the positive and negative impacts on the results are shown in the SHAP summary plot. Figure 4 shows the CNN-GRU summary plot of the model trained using date factor, meteorological factor and historical load factor. From the figure, it can be seen that when using sliding windows for load forecasting, in addition to the greatest influence of the load factor itself, in the date influence factor, the moment of the day (hourofday) and whether or not it is during working hours (work) have a great influence on the construction of the model. In the meteorological factors the average temperature (tem) of each day has a greater influence on the model compared to humidity (hum) and rainfall (mm). In the load factor the load at the same moment of the previous day (lag_1) has a greater effect than the load at the same moment of the previous seven days (lag_7).

It can be confirmed from Figure 5 that when the time of day is from near midnight to before dawn, the impact on the load is negatively correlated. There is a positive correlation between the influence of the time of day, that is, after 7 o'clock, on the

load. From Figure 6, it can also be confirmed that 0 represents non-working hours and 1 represents working hours. The influence on load during working hours is positively correlated, and some samples are negatively correlated because the working time of some samples is close to midnight. During non-working hours, the load effect was negatively correlated, while some samples were close to working hours, so the load effect was positively correlated.

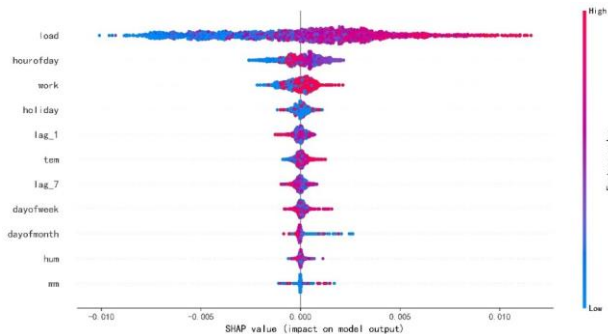


Fig.4 Summary and analysis of the overall characteristics

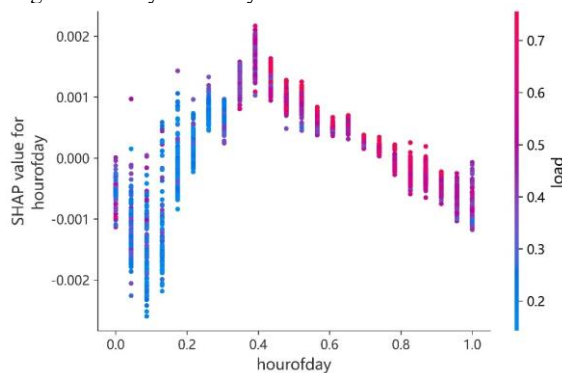


Fig.5 Time of day and load relationship

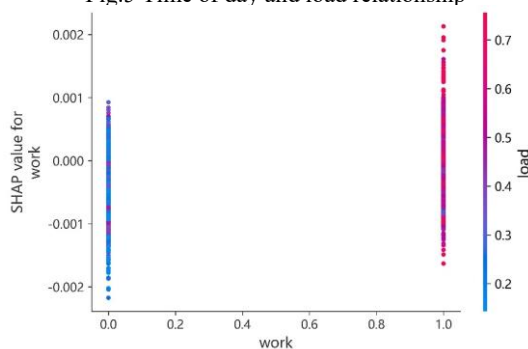


Fig.6 The relationship between working hours and load

From Figure 7, it can be estimated that when the temperature is higher than a certain value close to 1 (that is, summer), the impact on the load is positively correlated. When the temperature is low and close to 0 (that is, in winter), the impact on the load is negatively correlated.

V. CONCLUSIONS

In this paper, a methodology is proposed for electricity load forecasting. Firstly, data is collected and preprocessed for input

variables, and both external and internal factors are considered. Subsequently, the dataset is divided into training, validation, and test sets, and three baseline models—LSTM, GRU, CNN-LSTM—and a comparison model—CNN-GRU—are constructed using the training set. Using the test set, we compare their prediction performance in terms of MAPE, MAE, and RMSE. Finally, SHAP is applied to the dataset to interpret the CNN-GRU model.

The experimental analysis results show that the CNN-GRU hybrid network fully combines the advantages of CNN network and GRU network. It can not only train the model quickly, but also predict the results accurately, which has better performance compared with the existing methods. There are many influencing factors in load forecasting. In this paper, it is proved that among the date influencing factors, the working hour of the day especially during the daytime has a greater and positive impact on the load as compared to other date factors. Among the meteorological factors average temperature has a greater impact on load in summer than rainfall per day and humidity per day, which are also positively correlated.

One of the main limitations of this study is that the analysis only predicts hourly power consumption for one dataset. Other important information can be found when analyzing other types of datasets as well as other time intervals (e.g., 15- and 10-minute intervals). In addition to this, a hybrid forecasting model or an interpretable online learning model, which is based on two or more artificial intelligence techniques, should be developed that is capable of learning and reflecting recent trends and patterns of electricity consumption in order to improve the accuracy of load forecasting. In this study, the load model did not perform multi-step forecasting, and the CNN-GRU model needs to be improved to support multi-step forecasting to further improve the performance of the model.

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